

Ghali Ouaritini

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Education

The Pennsylvania State University
Bachelor of Science in Software Engineering (GPA: 3.24)

Graduated May 2025
Erie, PA

- Relevant Coursework: Data Structures and Algorithms (C++), Software Verification, Testing & Validation, Operating Systems, Object-Oriented Programming, Database Management System, Digital Embedded Systems, Embedded Real Time Systems, Machine Learning

Experience

Data Visualization Technician
Penn State Behrend – IT Department

Oct 2024 – Present

- Designed and developed interactive dashboards in **Power BI** to visualize key university data, including admissions trends, resource usage, and financial performance.
- Utilized **DAX functions**, **Excel**, and **database optimization techniques** to manage and manipulate data, enabling Penn State departments to derive valuable insights efficiently.
- Optimize reporting processes, reducing generation time by 30% and enhancing operational efficiency.
- Collaborated with departments across Penn State to deliver tailored visualizations that support institutional goals.

Machine Learning Developer
HD Forensics LLC

Sept 2024 – Present

- Developing a mobile application utilizing **machine learning** and **image processing** techniques, including feature extraction, and classification, to assist forensic professionals in identifying human bones in the field.
- Trained **convolutional neural networks (CNNs)** on 2D and 3D scans, achieving an accuracy of 90% in differentiating between human and animal bones.
- Collaborated with forensic experts at HD Forensics using **Agile methodologies** to develop a mobile app and ensure it meets client needs.

Projects

GrooveCircle Music Application | MongoDB, Node.js, Express.js, React, Spotify OAuth

- Architected and developed many microservices, including user authentication, playlist management, and activity feeds, to streamline system functionality.
- Collaborated with a team to plan and design **scalable APIs** for seamless communication between services.
- Integrated Spotify OAuth for secure user authentication and personalized music recommendations.
- Optimized **MongoDB** queries and API endpoints, reducing response times by 40% and enhancing overall application performance.

Stock Predictor Machine Learning System | Python, Scikit-learn, TensorFlow, Pandas, Matplotlib

- Designed and implemented predictive models including Random Forest, Support Vector Machines (SVM), and LSTM Neural Networks to forecast Apple stock trends.
- Performed extensive **data preprocessing**, including normalization and **feature engineering**, to enhance model accuracy.

Selenium Automated Testing Project | C#, .NET

- Created a user-friendly store management tool using Selenium and **Behavior-Driven Development (BDD)** to automate **GUI testing**, simplifying operations for non-technical users and ensuring software reliability.

Certifications & Involvement

ACM Club Hackathon | Penn State Behrend

April 2025

IEEE Embedded Systems Project Lead | IEEE Behrend Chapter

Jan 2025- Present

Fundamentals of Deep Learning Certificate | NVIDIA

Oct 2024

Software Development Organization | Member

Mar 2023 -Present

Math and Science, Tutor | Learning Resource Center

Sept 2021 - June 2022

Technical Skills

Front End | React, JavaScript, TypeScript, HTML, CSS, Bootstrap

Back End | Java, MongoDB, Node.js, Express.js, Python, C++, .NET, C#, SQL Server, RESTful API Design

Testing/Deployment | React Testing Library, Selenium, Docker, Postman, GitHub Actions, Streamlit, VMware

Developer Tools | AWS, Azure, Visual Studio, IntelliJ IDEA, JetBrains IDEs, GitHub, Cloud9, Linux, PowerBI

Hardware Tools | Verilog, Xilinx Vivado, Raspberry Pi 4, Assembly